

INDIVIDUAL PROJECT (LSDS)

WEEK 1 REPORT

(Problem Statement, Dataset Selection & Preprocessing)

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1) Domain Selection: Automotive Market Analytics (Finance Domain)

This project focuses on analyzing automobile pricing data to understand how different vehicle features affect market price trends.

2) Dataset Selection

- Dataset Name: Car Price Analysis Dataset
- Source: Kaggle
- Format: CSV
- Original Size: 156 KB
- Expanded Size: 152 MB
- Total Records After Expansion: ~1,000,000+ rows

3) Dataset Attributes:

- Car ID
- Brand
- Manufacturing Year
- Engine Size
- Fuel Type
- Transmission
- Mileage
- Condition

- Price
- Model

4) Problem Statement:

The objective of this project is to analyze large-scale automobile price data using the Hadoop ecosystem. The project aims to identify patterns and trends affecting car prices such as brand, fuel type, transmission type, manufacturing year, and mileage. The study demonstrates how distributed storage and processing using Hadoop can efficiently handle and analyze large datasets.

5) Objectives:

- To preprocess automobile pricing dataset.
- To expand the dataset to large-scale format suitable for Hadoop.
- To store data in HDFS.
- To perform distributed processing using MapReduce.
- To analyze trends using Hive and Pig.
- To derive meaningful business insights.

6) Data Preprocessing:

The following preprocessing steps were performed:

- Checked for missing values.
- Verified data types (numeric and categorical).
- Ensured consistent formatting.
- Removed inconsistencies if any.
- Expanded dataset using Python to simulate large-scale data (~152MB).
- Saved cleaned dataset as `car_price_big.csv`.

